MSG No. 59.405 $Pmmn$ [Type I, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[2,3,5,8]`

Table 3: Wyckoff site: `4c`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `-1`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{4}, y, z\right]$	`[1,6]`
2	$\left[\frac{3}{4}, -y, -z\right]$	`[2,5]`
3	$\left[\frac{3}{4}, y+\frac{1}{2}, -z\right]$	`[3,8]`
4	$\left[\frac{1}{4}, \frac{1}{2}-y, z\right]$	`[4,7]`

Table 6: Wyckoff site: `4f`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, z\right]$	`[1,7]`
2	$\left[x+\frac{1}{2}, \frac{3}{4}, -z\right]$	`[2,8]`
3	$\left[-x, \frac{3}{4}, -z\right]$	`[3,5]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	[4,6]

Table 7: Wyckoff site: 8g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[2]
3	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[6]
7	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]